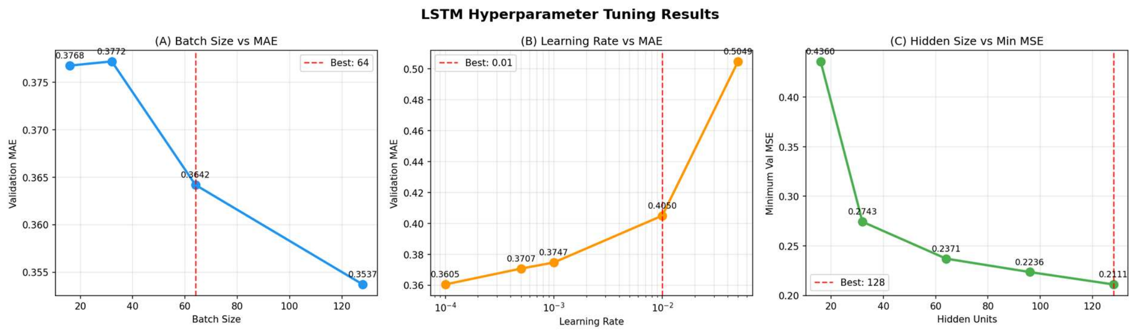
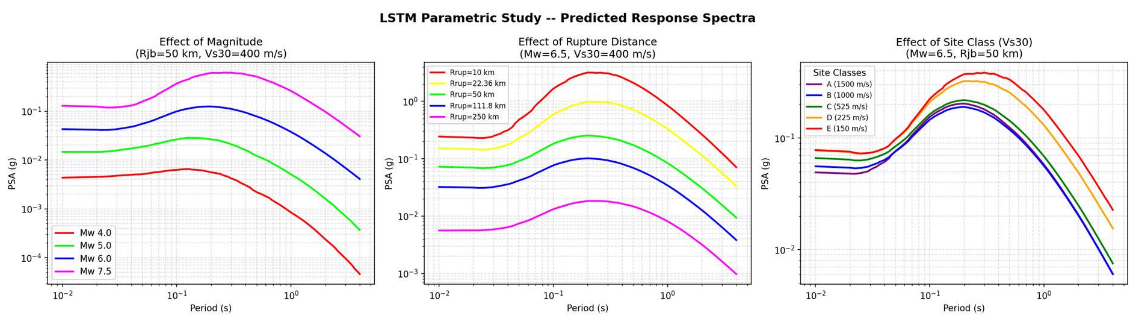
LSTM

1) Hyperparameter tuning

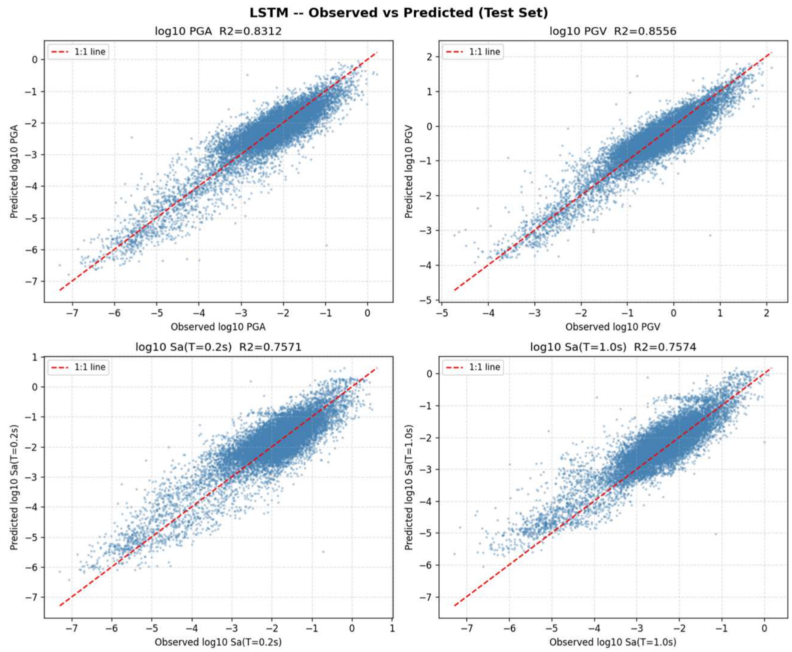


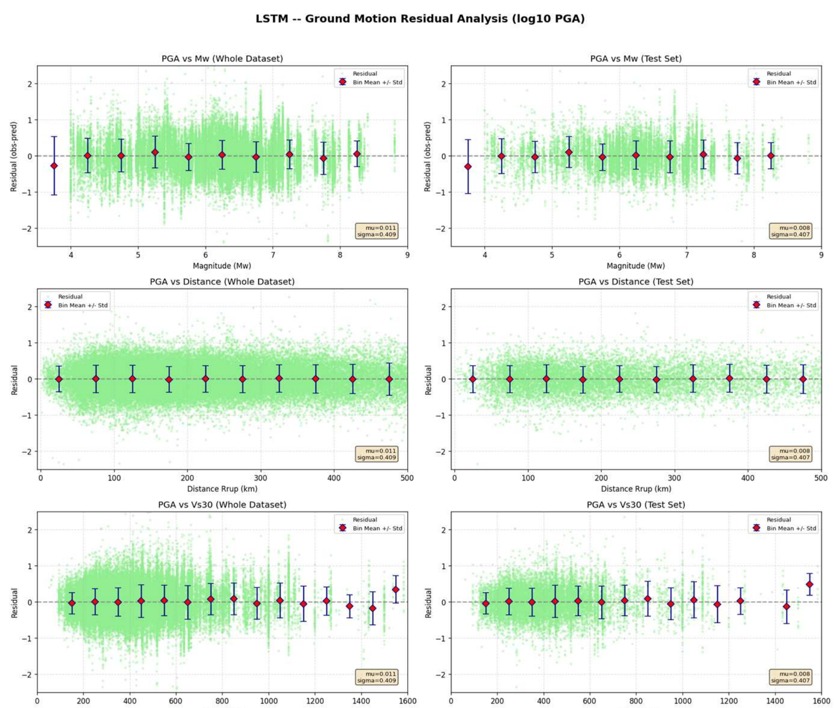
The hyperparameter tuning successfully identified the most efficient network settings. A batch size of 64 and a learning rate of 0.01 provided the fastest and most stable reduction in Mean Absolute Error (MAE). Evaluating the network capacity revealed that 128 hidden units resulted in the lowest Mean Squared Error (MSE), indicating that the LSTM required a relatively large memory state to capture the complex sequential patterns of the seismic data without overfitting.



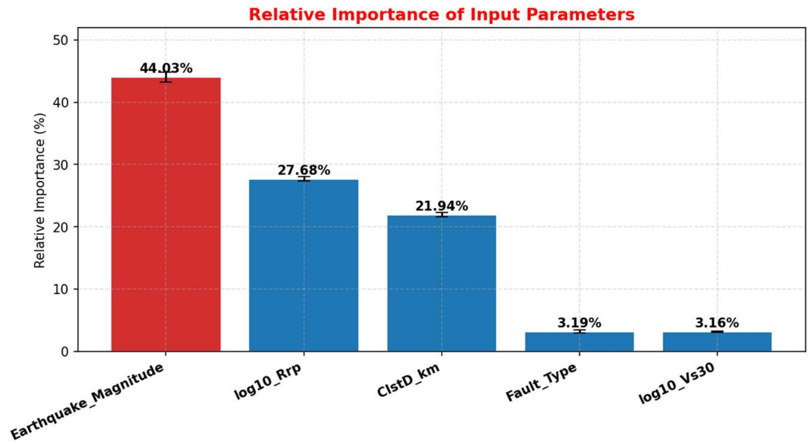
The parametric studies demonstrate that the LSTM model correctly learned the fundamental physics of earthquake engineering. The predicted response spectra organically behave exactly as physical laws dictate:

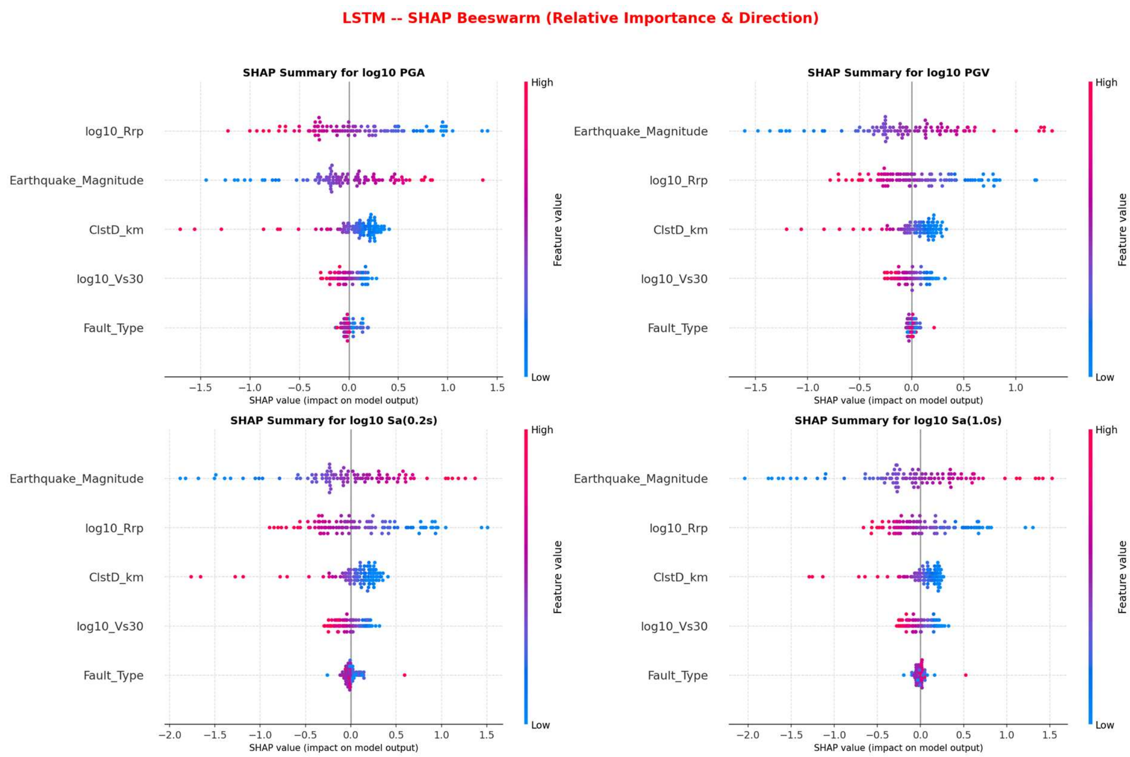
- **Magnitude:** The intensity of the spectra scales upward as earthquake magnitude increases.
- **Distance:** The spectra show clear wave attenuation, dropping in intensity as the rupture distance increases.
- **Site Condition:** The model captures soil amplification perfectly, predicting much higher spectral accelerations for softer soil profiles (Site Classes D and E) compared to hard rock (Site Class A).





The residual analysis proves the model is physically unbiased. When plotting the prediction errors (Observed – Predicted) against Magnitude (M_w), Rupture Distance (R_rup), and Site Condition (Vs_30), the data is evenly scattered around zero. The binned average markers sit perfectly on the zero line across all ranges, indicating the model does not systematically over-predict or under-predict shaking for specific earthquake sizes, distances, or soil types.





The feature importance analysis quantifies the driving forces behind the model's predictions. Earthquake Magnitude (44%) and Rupture Distance (27%) are the overwhelmingly dominant inputs. The SHAP beeswarm plots visualize the physical logic the model learned: higher magnitudes (red dots) strongly push predictions higher, while larger distances (red dots) strongly push predictions lower. This confirms the model relies on the correct, real-world laws of seismic energy release and attenuation to make its forecasts.